SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	**CSA1223**	**Course Title:**	**Computer Architecture**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
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Department:	AIML	**Date:**	24/8/2026

84%
48

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

1. Smartphone Performance Optimization
   A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

2. Cloud Server Memory Bottleneck
   A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

3. Autonomous Vehicle Data Processing
   An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

4. AI Inference Edge Device
   An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

5. High-Performance Gaming Console
   A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

Code of Conduct:

I Diwakar S 19xxxxxx *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S Diwakar

MARKS ALLOCATION

	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1	1	1.25	1	0.75	0.75
Q2	1	1.25	1	0.5	0.5
Q3	0.5	1	1.25	0.5	0.5
Q4	0.5	1	1.25	0.75	0.75
Q5	1	1.25	1	0.5	0.5

21.25

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints(1)	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem(1.25)	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards(1.25)	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data(0.75)	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation(0.5)	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

1.Smartphone Performance Optimization .A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.

Smartphone Performance Optimization

Problem

A mobile manufacturer observes lag while switching between apps. The main goal is to reduce memory access latency and improve system throughput.

Answer

When a smartphone switches between applications, the processor needs to access instructions and data from different memory levels. If the required data is not available in the faster cache, the CPU must access slower memory such as DRAM. This causes delays and app-switching lag.

Role of L1, L2, L3 Cache and DRAM

L1 Cache:

• L1 is the smallest and fastest cache.

• It is located very close to or inside each CPU core.

• It stores frequently used instructions and data.

• It provides very low access latency.

• Its capacity is limited.

L2 Cache:

• L2 is larger than L1 but slightly slower.

• It stores data that cannot fit in L1.

• It reduces accesses to slower memory.

• It acts as an intermediate level between L1 and L3/DRAM.

L3 Cache:

• L3 is larger than L1 and L2.

• It is generally shared among multiple CPU cores.

• It stores commonly accessed data and instructions.

• It reduces expensive DRAM accesses.

• It improves performance when several applications are running.

DRAM:

• DRAM is the main memory of the smartphone.

• It has much greater capacity than cache.

• It is slower than L1, L2 and L3.

• Applications and their working data are stored in DRAM while running.

• Frequent DRAM accesses can increase latency and power consumption.

Proposed Hierarchical Memory Redesign

CPU → L1 Cache → L2 Cache → L3 Cache → DRAM → Storage

The manufacturer can improve performance by:

1. Increasing L1 cache moderately for frequently used data.

2. Using a sufficiently large L2 cache to reduce L3 accesses.

3. Increasing L3 cache to hold shared application data.

4. Using high-bandwidth, low-latency DRAM.

5. Improving cache replacement and prefetching techniques.

6. Keeping frequently used application data in cache.

7. Reducing unnecessary movement of data between DRAM and cache.

Result

The redesigned hierarchy reduces the number of slow DRAM accesses. Therefore, application switching becomes faster, CPU waiting time decreases, and overall smartphone throughput improves.

2.Cloud Server Memory Bottleneck. A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.

Cloud Server Memory Bottleneck

Problem

A cloud service provider experiences high latency during database queries. The problem requires comparison between cache hierarchy and virtual memory paging.

Answer

Database servers perform a large number of memory accesses. High query latency can occur when frequently accessed data is not available in cache or when the operating system frequently performs virtual-memory paging.

Cache Hierarchy

L1 – Very high speed, small capacity, frequently used data.

L2 – High speed, medium capacity, backup for L1.

L3 – Moderate speed, large capacity, shared data.

DRAM – Lower speed than cache, very large capacity, main memory.

Virtual Memory Paging

Virtual memory allows the operating system to use storage as an extension of physical memory. When physical memory is insufficient, pages may be moved from RAM to storage and later brought back. Storage is much slower than RAM, so excessive paging can produce high database latency.

Comparison

Cache hierarchy:

• Works mainly at the processor-memory level.

• Uses small and very fast memory.

• Reduces CPU memory-access latency.

• Useful for frequently accessed data.

Virtual memory paging:

• Managed by the operating system.

• Provides a larger logical memory space.

• Uses storage when RAM is insufficient.

• Excessive paging increases latency.

Recommended Improvement

1. Increase available DRAM for database workloads.

2. Use larger and efficient L3 cache where supported.

3. Use high-performance DRAM.

4. Reduce unnecessary paging.

5. Optimize database memory allocation.

6. Keep frequently accessed database pages in RAM.

7. Use fast SSD-based storage for unavoidable paging.

8. Improve application and database caching.

Result

The combination of an efficient cache hierarchy and sufficient DRAM reduces CPU and memory delays. Minimizing paging prevents slow storage accesses and improves database query response time.

3.Autonomous Vehicle Data Processing. An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.

3. Autonomous Vehicle Data Processing

Problem

An autonomous vehicle must process sensor data in real time. The memory system must provide high bandwidth and low latency.

Answer

An autonomous vehicle receives data continuously from cameras, radar, LiDAR, GPS and other sensors. The processor must process this information quickly to make decisions. If memory access is slow, the vehicle's processing system may not respond quickly enough.

SRAM

SRAM is very fast, has low latency, is more expensive per bit, and is usually used for CPU caches and small high-speed buffers. It is suitable for critical real-time data and frequently accessed instructions.

DRAM

DRAM has higher capacity than SRAM, is slower than SRAM, is less expensive per bit, and is suitable for storing large amounts of sensor and application data. It can serve as the main working memory.

MRAM

MRAM is a non-volatile memory technology. It retains data without continuous power, can provide fast access compared with many conventional non-volatile memories, has good endurance, and can be useful for persistent data and fast-access storage.

Proposed Memory Hierarchy

CPU/AI Processor

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SRAM Cache

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High-bandwidth DRAM

↓

MRAM

↓

Long-term Storage

Working

1. SRAM stores the most frequently accessed sensor-processing data.

2. DRAM stores large real-time datasets and active application information.

3. MRAM stores important persistent information and data that should remain available without continuous power.

4. Long-term storage stores large datasets that are not immediately required.

Result

This hierarchy provides low latency through SRAM, high capacity and bandwidth through DRAM, and non-volatility through MRAM. Therefore, the autonomous vehicle can process sensor information efficiently in real time.

4.AI Inference Edge Device An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.

AI Inference Edge Device

Problem

An edge AI device requires fast model loading while maintaining low power consumption. The question requires comparison of NAND Flash, DRAM and RRAM.

Answer

Edge AI devices such as smart cameras and IoT devices need to load AI models quickly. At the same time, they often have limited battery power. Therefore, the memory system should provide a balance between speed, capacity, power consumption and cost.

NAND Flash

NAND Flash is non-volatile, has high capacity, is relatively low cost, and is suitable for permanent storage. It is slower than DRAM for active processing. It can store AI models, applications and data.

DRAM

DRAM is faster than NAND Flash, is volatile, is suitable for active AI model execution, and generally consumes more power than some non-volatile technologies. It is used for loading and running the AI model.

RRAM

RRAM (Resistive RAM) is non-volatile, can provide fast access, has potential for low-power operation, and can be useful for emerging AI-memory applications.

Proposed Hierarchical Structure

AI Processor

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Fast Cache/SRAM

↓

DRAM

↓

RRAM

↓

NAND Flash

Working

1. AI models are permanently stored in NAND Flash.

2. Frequently used model parameters can be kept in RRAM.

3. The active model is loaded into DRAM.

4. The processor accesses the most frequently required information from fast cache/SRAM.

5. Unused data remains in lower memory levels.

Optimization

• Avoid unnecessary data movement.

• Load only required model components.

• Use model compression where appropriate.

• Keep frequently used data in faster memory.

• Use RRAM for suitable non-volatile workloads.

• Reduce unnecessary DRAM accesses.

Result

The proposed hierarchy provides large storage through NAND Flash, fast working memory through DRAM, and low-power non-volatile storage through RRAM. This helps the edge AI device load models quickly while controlling power consumption.

5.High-Performance Gaming Console. A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

5. High-Performance Gaming Console

Problem

A gaming console experiences frame drops while loading large game scenes. The solution requires analysis of L3 cache and DRAM throughput.

Answer

Modern games contain large textures, 3D models, lighting information, animations and other assets. When a new scene is loaded, large amounts of data must move between storage, DRAM and the processor. If the processor cannot receive data quickly enough, frame generation may be delayed, resulting in frame drops.

Role of L3 Cache

L3 cache is faster than DRAM, has larger capacity than L1 and L2, can store frequently accessed game data, reduces the number of accesses to DRAM, and helps CPU cores access shared data efficiently. However, its capacity is limited, so it cannot hold an entire large game scene.

Role of DRAM

DRAM has much greater capacity than L3 cache, stores large game assets and working data, and provides the main memory for the game. It has higher latency than cache. For large scene loading, DRAM bandwidth is particularly important because large amounts of data must be transferred.

L3 Cache vs DRAM

L3 Cache:

• Faster

• Smaller capacity

• Higher cost per bit

• Stores frequently used data

• Reduces CPU access latency

DRAM:

• Slower than cache

• Much larger capacity

• Lower cost per bit

• Stores large working datasets

• Provides high-capacity main memory

Proposed Memory Hierarchy

CPU → L1 → L2 → L3 → High-bandwidth DRAM → Storage

Improvements

1. Increase L3 cache capacity where practical.

2. Use high-bandwidth DRAM.

3. Improve memory controllers.

4. Use efficient data prefetching.

5. Keep frequently accessed game data in cache.

6. Reduce unnecessary memory transfers.

7. Stream large assets efficiently from storage to DRAM.

8. Optimize game-engine memory management.

Result

L3 cache reduces latency for frequently accessed data, while high-bandwidth DRAM handles large game scenes. Combining both provides faster scene loading and helps maintain stable frame rates.

Overall Conclusion

The five industry problems demonstrate the importance of hierarchical memory systems. A general hierarchy is: L1 Cache → L2 Cache → L3 Cache → SRAM/DRAM → Advanced Non-Volatile Memory → Storage. As we move toward lower levels, capacity generally increases while speed decreases. The main design principle is to keep frequently accessed and time-critical data in faster memory, while storing large amounts of less frequently accessed data in higher-capacity memory. This reduces latency and improves overall system throughput.